

Questions

Set A:

A smart fan initially having “low” speed and 24 temperature changes its speed to the next speed when the nextSpeed() method is called. There are 3 speeds in total - low, medium and high.

The coolDown() method will decrease the temperature according to the speed as follows:

- If the current speed is “high”, temperature will decrease by 5.
- If the current speed is “medium”, temperature will decrease by 3.
- If the current speed is “low”, temperature will decrease by 1.

If decreasing the temperature makes it go below 18, it should be kept as 18.

Write the design class following the given tester code and output.

Tester Code	Output
<pre> public class Tester { public static void main(String[] args) { SmartFan fan1 = new SmartFan(); System.out.println("1====="); fan1.info(); System.out.println("2====="); fan1.nextSpeed(); System.out.println("3====="); System.out.println(fan1.changeTemperature(4)); System.out.println("4====="); fan1.info(); System.out.println("5====="); fan1.coolDown(); System.out.println("6====="); fan1.nextSpeed(); fan1.coolDown(); System.out.println("7====="); fan1.nextSpeed(); fan1.info(); System.out.println("8====="); fan1.coolDown(); fan1.nextSpeed(); fan1.coolDown(); System.out.println("9====="); fan1.info(); fan1.coolDown(); } } </pre>	<pre> 1===== Current Speed: Low Current Temperature: 24 2===== Speed changed to: Medium 3===== Temperature set to: 28 4===== Current Speed: Medium Current Temperature: 28 5===== Temperature after cooling: 25 6===== Speed changed to: High Temperature after cooling: 20 7===== Speed changed to: Low Current Speed: Low Current Temperature: 20 8===== Temperature after cooling: 19 Speed changed to: Medium Temperature after cooling: 18 9===== Current Speed: Medium Current Temperature: 18 Temperature after cooling: 18 </pre>

Set B:

A camera, initially with "Photo" mode and 70% battery percentage changes to the next mode when the nextMode() method is called. There are 3 modes in total: Photo, Portrait, and Video.

The capture() method decreases the battery according to the current mode as follows:

- If the current mode is "Video", battery decreases by 20.
- If the current mode is "Portrait", battery decreases by 10.
- If the current mode is "Photo", battery decreases by 5.

If decreasing the battery percentage makes it go below 40, it should be kept as 40.

Write the design class following the given tester code and output.

Tester Code	Output
<pre>public class Tester { public static void main(String[] args) { Camera camera1 = new Camera(); System.out.println("1====="); camera1.info(); System.out.println("2====="); camera1.nextMode(); System.out.println("3====="); System.out.println(camera1.chargeBattery(10)); System.out.println("4====="); camera1.info(); System.out.println("5====="); camera1.capture(); System.out.println("6====="); camera1.nextMode(); camera1.capture(); System.out.println("7====="); camera1.nextMode(); camera1.info(); System.out.println("8====="); camera1.capture(); camera1.nextMode(); camera1.capture(); System.out.println("9====="); camera1.info(); camera1.capture(); } }</pre>	<pre>1===== Current Mode: Photo Current Battery: 70 2===== Mode changed to: Portrait 3===== Battery charged to: 80 4===== Current Mode: Portrait Current Battery: 80 5===== Battery after capture: 70 6===== Mode changed to: Video Battery after capture: 50 7===== Mode changed to: Photo Current Mode: Photo Current Battery: 50 8===== Battery after capture: 45 Mode changed to: Portrait Battery after capture: 40 9===== Current Mode: Portrait Current Battery: 40 Battery after capture: 40</pre>

Solutions

Set A:

```
class SmartFan {
    String speed= "Low";
    int temperature =24;
    public void info() {
        System.out.println("Current Speed: " + speed);
        System.out.println("Current Temperature: " + temperature);
    }
    public void nextSpeed() {
        if(speed.equals("Low")) {
            speed = "Medium";
        }
        else if(speed.equals("Medium")) {
            speed = "High";
        }
        else{
            speed = "Low";
        }
        System.out.println("Speed changed to: " + speed);
    }
    public String changeTemperature(int value) {
        temperature += value;
        return "Temperature set to: " + temperature;
    }
    public void coolDown() {
        if(speed.equals("Low")) {
            temperature -= 1;
        }
        else if(speed.equals("Medium")) {
            temperature -= 3;
        }
        else{
            temperature -= 5;
        }
        if (temperature < 18) {
            temperature = 18;
        }
        System.out.println("Temperature after cooling: " + temperature);
    }
}
```

Set B:

```
public class Camera {
    String mode= "Photo";
    int battery= 70;
    public void info() {
        System.out.println("Current Mode: " + mode);
        System.out.println("Current Battery: " + battery);
    }
    public void nextMode() {
        if (mode.equals("Photo")) {
            mode = "Portrait";
        }
        else if (mode.equals("Portrait")) {
            mode = "Video";
        }
        else {
            mode = "Photo";
        }
        System.out.println("Mode changed to: " + mode);
    }
    public String chargeBattery(int value) {
        battery += value;
        return "Battery charged to: " + battery;
    }
    public void capture() {
        if (mode.equals("Video")) {
            battery -= 20;
        }
        else if (mode.equals("Portrait")) {
            battery -= 10;
        }
        else {
            battery -= 5;
        }

        if (battery < 40) {
            battery = 40;
        }
        System.out.println("Battery after capture: " + battery);
    }
}
```

Rubric

For both sets:

Criteria	Points
Class name and instance variables	1.5
next() method - 3 conditions + output	2
changeTemperature() / chargeBattery() method - proper update and return statement	2
coolDown() / capture() method - proper updates and output	3.5
info()	1
Total:	10